Wisdom is All You Need: Learning to Sequentially Invert Reasoning Traces is a Free Unsupervised Reward Signal

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Abstract

Current Large Language Models (LLMs) are fine-tuned for reasoning via next-token prediction, a process that masters sequential, forward-pass logic. While powerful, this approach results in a brittle, procedural understanding, akin to memorizing a path rather than learning a map. We introduce a novel, self-supervised fine-tuning methodology called Bidirectional Consistency (BiCo). The method augments standard '(question -¿ answer)' training examples with a second, self-supervised task: given the original question, the model must generate the entire reasoning trace in a temporally inverted token sequence. Training on this mixed dataset with a standard Supervised Fine-Tuning (SFT) objective compels the model to discard superficial, directional statistical correlations in favor of a more abstract, robust, and symmetrical internal representation of knowledge. We term the emergent property of this representation "wisdom"—a shift from procedural recitation to structural understanding. Crucially, this method requires no new human-labeled data. It allows 100% of logged forward-pass inference data to be recycled into training examples, creating a perpetual, zero-cost data flywheel for model improvement. This provides a computationally efficient and infinitely scalable alternative to reinforcement learning for deepening a model's reasoning capabilities. Our empirical validation on the GSM8K benchmark shows that a 0.5B parameter model fine-tuned with BiCo significantly outperforms the same model trained with standard forward-only SFT, demonstrating the effectiveness of this approach.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating coherent, multi-step reasoning traces [4]. These successes are founded on the autoregressive, next-token prediction paradigm, which trains models to sequentially complete a text prompt. This process excels at learning directional, causal, and narrative patterns, effectively modeling $P(\text{answer}|\text{question})$.

However, this unidirectional training has fundamental limitations. The model's knowledge is often procedural rather than structural. To use an analogy, a model can learn to recite the alphabet forwards perfectly, but this does not guarantee it understands the alphabet as an ordered set. It may fail when asked to recite it backwards, or struggle with a query requiring random access, such as "What is the 13th letter?". This brittleness manifests in LLMs as logical inconsistencies, susceptibility to superficial prompt phrasing, and a general lack of robust understanding [5].

Current methods to improve reasoning, such as self-consistency [3] and Reinforcement Learning from Human Feedback (RLHF) [1], rely on generating multiple outputs at inference time or require

expensive, human-curated preference datasets. Both approaches treat reasoning as a process to be improved at the sequence level, rather than improving the underlying knowledge representation within the model's weights.

In this work, we propose a novel training methodology, Bidirectional Consistency (BiCo), designed to forge a deeper, more structural form of knowledge directly into the model's parameters. We hypothesize that true understanding, or "wisdom," is characterized by the bidirectionality of knowledge representations. If a model truly understands the relationship between a premise Q and a conclusion A, its internal representation should be robust to the temporal direction of generation. By forcing a single model to generate both a forward trace and a temporally-reversed trace from the same starting point, we create an optimization pressure that favors the learning of abstract, symmetrical concepts over shallow, directional statistics.

Our central contribution is to show that this objective can be optimized using a simple, self-supervised fine-tuning process. Every user interaction that generates a reasoning trace can be programmatically inverted and recycled as a training example. This creates a zero-cost, infinitely scalable data flywheel, turning the model's inference compute into a source of high-quality training data for its own improvement.

2 Methodology

Our approach does not introduce a new loss function but rather a novel data augmentation strategy used with a standard Supervised Fine-Tuning (SFT) objective. Let an LLM be a function F_θ with parameters θ. A reasoning trace is a sequence of tokens $T = (t_1, ..., t_n)$.

Definition 1 (Forward Training Example)**.** *Given a question or premise Q and its corresponding reasoning and answer A, the full forward trace is the sequence $S_{fwd} = [Q; A]$. In a standard SFT setup, the model is trained to predict the tokens of A conditioned on Q. The loss is the standard cross-entropy over the answer sequence.*

$$\mathcal{L}_{SFT}(S_{fwd}) = -\sum_{i=1}^{|A|} \log P(a_i | Q, a_1, ..., a_{i-1}; \theta)$$

Definition 2 (Backward Training Example)**.** *Let $S_{fwd} = (t_1, ..., t_n)$ be the full forward trace including the question and answer. We define a temporal reversal function $rev(S) = (t_n, ..., t_1)$. The backward training example consists of the original question Q as the prompt, and the temporally reversed full trace $rev(S_{fwd})$ as the target sequence. Special tokens may be used to differentiate the task, e.g., $S_{bwd} = [¡backwards¿, Q, ¡/backwards¿; rev(S_{fwd})]$. The model's task is to autoregressively generate the reversed sequence.*

$$\mathcal{L}_{SFT}(S_{bwd}) = -\sum_{i=1}^{n} \log P(t_{n-i+1} | Q, t_n, ..., t_{n-i+2}; \theta)$$

The BiCo training objective is simply the standard SFT loss applied to a dataset that is a union of both forward and backward examples.

Definition 3 (Bidirectional Consistency (BiCo) Training)**.** *Given a dataset of forward traces $\mathcal{D}_{fwd} = \{(Q_j, A_j)\}$, we construct an augmented dataset $\mathcal{D}_{BiCo} = \mathcal{D}_{fwd} \cup \mathcal{D}_{bwd}$, where $\mathcal{D}_{bwd}$ contains the corresponding backward examples. The model is then fine-tuned by minimizing the standard cross-entropy loss over this combined dataset.*

$$\mathcal{L}_{BiCo} = \mathbb{E}_{(X,Y)\sim\mathcal{D}_{BiCo}}[-\log P(Y|X; \theta)]$$

Remark 1. *This is a self-supervised objective. The backward examples are synthesized programmatically from the forward examples, requiring no external labels. This makes it ideal for SFT using existing model inference logs.*

3 From Procedural Knowledge to Structural Wisdom

The BiCo objective creates a powerful inductive bias. Standard SFT on forward traces encourages the model to learn a brittle, procedural representation, akin to a singly-linked list where each concept points only to the next. The backward training task directly penalizes this structure.

Proposition 1 (The Wisdom Hypothesis)**.** *Minimizing the BiCo objective forces the model's parameter optimization to converge on internal representations that are abstract and symmetrical. An autoregressive model cannot solve the temporal reversal task by simply following a logical procedure; it must have a holistic representation of the entire target sequence before generation begins. The optimizer's most efficient solution is to form representations that encode the fundamental, non-directional relationship between concepts. This structural knowledge is what we define as wisdom.*

Consider the alphabet analogy.

1. **Procedural Knowledge (Standard SFT):** The model learns $P('B'|'A')$, $P('C'|'A',' B')$, etc. The concept ''B'' is primarily encoded as "what comes after 'A'".

2. **Bidirectional Constraint (BiCo Training):** The model must also learn to generate the sequence 'Z, Y, X...' given a prompt. To do this, it learns $P('Z'|\text{prompt})$, $P('Y'|\text{prompt},' Z')$, etc. The model cannot simply follow the 'A-¿B' chain.

3. **Structural Knowledge (Wisdom):** To solve both tasks efficiently, the model is incentivized to form a single, abstract representation where the relationships between letters are symmetrical. This robust internal model allows for emergent capabilities, such as random access ("What is the 3rd letter?")—a query that is difficult to answer with purely sequential knowledge.

By training with the BiCo objective, we are not teaching the model to "reason backwards" in a logical sense. We are using the temporal reversal task as a tool to forge a fundamentally different and more powerful knowledge structure within the model's weights. The goal is to make the model's standard forward-pass inference more robust, as it will now emanate from this wiser, structural representation.

4 The Self-Supervised Data Flywheel

The most significant practical advantage of the BiCo objective is its compatibility with existing data pipelines. It transforms every piece of inference data into a rich training signal at zero marginal cost.

We propose the following continuous fine-tuning loop:

1. **Serve & Log:** A deployed LLM F_{θ_t} serves user requests. All interactions that produce a reasoning trace (Q, A) are logged.

2. **Synthesize & Recycle:** A simple, automated script processes these logs. For each log (Q, A), it creates a pair of structured data entries: one for the forward task and one for the backward (temporally inverted) task.

3. **Fine-Tune & Forge:** The model is periodically fine-tuned on this dataset of recycled traces using the standard SFT loss function. This SFT run produces an improved model $F_{\theta_{t+1}}$.

4. **Deploy & Improve:** The new model $F_{\theta_{t+1}}$ is deployed. Because its internal representations are more robust, it generates slightly higher-quality, more coherent reasoning traces, which in turn provide better data for the next cycle of the flywheel.

This creates a virtuous cycle of self-improvement. The system's scalability is limited only by its inference capacity, and the training data naturally reflects the distribution of tasks users find valuable. This stands in stark contrast to RLHF, which faces a bottleneck in the acquisition of expensive and slow human preference data.

5 Empirical Validation

We conducted an experiment on the GSM8K benchmark using the 'Qwen/Qwen1.5-0.5B' model. We compared three conditions: the base model's zero-shot performance, the performance after standard forward-only fine-tuning ('FwdCo'), and the performance after fine-tuning with our BiCo methodology. The results show that while 'FwdCo' improves performance from 2.0

6 Implications and Future Work

We believe the BiCo training paradigm has several profound implications for the development of advanced AI.

- **Reduced Hallucination:** By enforcing a representation robust to temporal inversion, the model is less likely to produce conclusions that are not structurally supported by their premises.

- **Improved Zero-Shot Generalization:** Models trained with BiCo should develop more abstract concept representations, allowing them to apply learned logic to novel domains more effectively.

- **A Path to More Reliable AI:** This method provides a scalable mechanism for improving model reliability without constant human supervision, representing a step towards more trustworthy AI systems.

Future work will involve applying BiCo fine-tuning to a range of larger models on more complex reasoning benchmarks (e.g., MATH, MMLU). Further research will also explore the optimal ratio of forward-to-backward examples in the training mix and the impact of different temporal reversal strategies.

7 Related Work

Our work builds upon several key areas of LLM research.

Self-Consistency and Self-Correction: Wang et al. [3] showed that generating multiple reasoning paths at inference time and taking a majority vote improves performance. Our approach differs by aiming to bake this consistency directly into the model's weights during training, rather than relying on an expensive inference-time procedure.

Data Augmentation: BiCo is a form of highly structured, model-centric data augmentation. Unlike simple back-translation, it creates a direct consistency pressure on the model's internal representation of a complete thought process.

Knowledge Distillation: The process of continually fine-tuning the model on its own outputs resembles self-distillation. BiCo provides a novel objective for this process, focused on deepening understanding rather than merely compressing knowledge.

SFT vs. RLHF: While RLHF [2] has been highly effective, it is data-hungry and can lead to reward hacking. BiCo provides a complementary, SFT-based approach that is unsupervised and focuses on internal consistency rather than external preference matching.

8 Conclusion

We have introduced Bidirectional Consistency (BiCo), a self-supervised training methodology designed to transform the nature of knowledge within LLMs. By augmenting standard training with a task requiring the generation of temporally-inverted reasoning traces, we create a powerful pressure for the model to move beyond shallow procedural learning towards a deep, structural understanding we call wisdom.

The primary advantage of this method is its perfect alignment with the operational realities of deploying large-scale AI. It turns the data exhaust of inference into the fuel for self-improvement, creating a zero-cost, infinitely scalable flywheel for enhancing model reasoning. BiCo offers a promising and pragmatic path toward building more robust, reliable, and genuinely intelligent AI systems.

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